

Experimental Setup

Are the operator and observer identities consistent under weighted nonunitary frame transformations?

Experiment: gross.operator-identities.v1 Model: signal-space.operator-constitution.v1

Runtime configuration: fixtures/research/gross-test-01.json

Plan lock: 16b1f2c194709efcbeae9427f137d388014e940239965b4fd8ac2689876218ae

MODEL AND NUMERICAL SETUP

Action: No evolution action; Hermitian operator constitution only, prerequisite shared by SS OPS 1 and SS OCF 1.

Units: Matrices, state weights and residuals dimensionless; near-collinearity angles radians.

Domain: 1000 random moderate-condition samples, tetrahedral projectors, fixed boost control and 10 near/singular aggregates.

Boundaries: No spatial domain or time integration. Conditioning, not mesh/domain/timestep, is the numerical control.

Initial Data: Root seed 2026092501; PCG64 with runtime sampling stream. Standard-normal Pauli coefficients and complex constituents; condition ≤ 100 ; independent SU(2)-boost-SU(2), rapidity in $[-1,1]$.

Solver: Registered direct-matrix-v1 batch with independent saved-input Pauli-component analysis; single-thread BLAS.

Tolerances: Scale-normalized valid-domain residual $< 1e-10$; positivity > 0 ; negative omitted-weight error $> 1e-3$. No hidden singular-value regularization.

COMPETING PREDICTIONS

consistent: Valid samples obey identities, exact Gram spectrum and observer positivity; singular and broken-weight controls are detected.

inconsistent: One or more valid-domain identities exceed threshold, positivity fails, or invalid-domain/weight controls escape detection.

CONTROLS

independent-components: Detect matrix-algebra and saved-output errors; method: Read all raw matrices and compare worker scalars with independent real component formulas.

tetrahedron: Check the exact algebraic Lorentzian signature; method: Compare computed Gram matrix eigenvalues to $(-1/3, -1/3, -1/3, 1)$.

singular-and-weight: Expose invalid observer domains and omitted boost weights; method: Rank-one and near-collinear aggregates plus a deterministic rapidity-one boost with and without weight transformation.

PREREGISTERED DECISIONS

coverage (All seeded samples and declared controls are saved and independently checked)

Pass: All seeded samples and declared controls are saved and independently checked

Fail: Complete valid evidence violates the specified condition.

Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.

Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

identities (All scale-normalized well-conditioned residuals are strictly below $1e-10$)

Pass: All scale-normalized well-conditioned residuals are strictly below $1e-10$

Fail: Complete valid evidence violates the specified condition.

Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.

Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

positive-observer (All valid observer forms and norms are positive)

Experimental Setup

Pass: All valid observer forms and norms are positive
Fail: Complete valid evidence violates the specified condition.
Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.
Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

tetrahedral (Gram spectrum is $(-1/3, -1/3, -1/3, 1)$ within $1e-10$)
Pass: Gram spectrum is $(-1/3, -1/3, -1/3, 1)$ within $1e-10$
Fail: Complete valid evidence violates the specified condition.
Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.
Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

weight-control (Correct weights pass; deliberately omitted weights give error above $1e-3$)
Pass: Correct weights pass; deliberately omitted weights give error above $1e-3$
Fail: Complete valid evidence violates the specified condition.
Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.
Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

singular-domain (Rank-one aggregate is rejected, near-collinear samples classified without regularization)
Pass: Rank-one aggregate is rejected, near-collinear samples classified without regularization
Fail: Complete valid evidence violates the specified condition.
Unresolved: Required raw evidence is incomplete or unavailable; no scientific pass may be issued.
Uncertainty: Floating-point arithmetic and declared condition domain, not a statistical confidence level.

VISUAL QUESTIONS

operator-residuals [evidentiary]: Do operator and observer identities remain accurate across the declared conditioning domain?

View: log residual versus conditioning scatter with frozen threshold; observables: condition_j, max_residual, norm_transform_error

consistent: Valid residuals below threshold, exact signature and correctly detected failure controls.

inconsistent: Threshold violation, wrong signature or an undetected failure control.

Controls: independent-components; uncertainty: Finite precision; no fitted error bars. Singular cases outside the accepted domain remain visible.

Data: reports/<report-id>/plot-data/residuals.csv; transforms: No smoothing or fitting; log residual plots display $\max(\text{error}, 1e-18)$, retaining original zeros.

tetrahedral-gram [evidentiary]: Do four null projector directions span the predicted Lorentzian operator space?

View: Gram heatmap and computed/exact eigenvalues; observables: gram, gram_eigenvalues

consistent: Valid residuals below threshold, exact signature and correctly detected failure controls.

inconsistent: Threshold violation, wrong signature or an undetected failure control.

Controls: tetrahedron; uncertainty: Finite precision; no fitted error bars. Singular cases outside the accepted domain remain visible.

Data: reports/<report-id>/plot-data/controls.json; transforms: No smoothing or fitting; log residual plots display $\max(\text{error}, 1e-18)$, retaining original zeros.

observer-domain [evidentiary]: Do singular aggregates and omitted ray weights expose the intended failure controls?

View: conditioning by angle, explicit singular rejection and correct/broken weight comparison;

observables: angle, condition_j, status, correct_weight_error, omitted_weight_error

Experimental Setup

consistent: Valid residuals below threshold, exact signature and correctly detected failure controls.
inconsistent: Threshold violation, wrong signature or an undetected failure control.
Controls: singular-and-weight; uncertainty: Finite precision; no fitted error bars. Singular cases outside the accepted domain remain visible.
Data: reports/<report-id>/plot-data/conditioning.json, reports/<report-id>/plot-data/controls.json;
transforms: No smoothing or fitting; log residual plots display $\max(\text{error}, 1e-18)$, retaining original zeros.

RESOURCE LIMITS

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{"max_memory_mb": 512, "max_output_mb": 16, "max_wall_seconds": 60}
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PROHIBITED INFERENCES

This is algebra/implementation validation, not physical Lorentz covariance, propagation, emergent spacetime or a clock.

No dynamics means conservation, boundary sensitivity and discretization convergence are not evaluated. Finite seeds do not prove identities or justify ill-conditioned observer normalization.